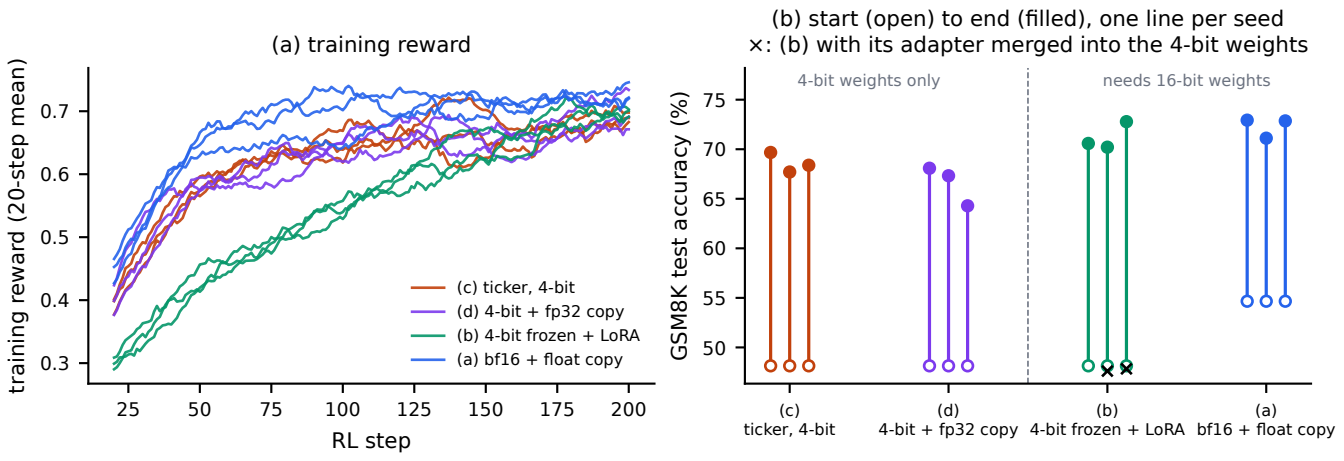
(a) training reward
training reward (20-step mean)
0.3
0.4
0.5
0.6
0.7
25
50
75
100
125
150
175
200
RL step
(c) ticker, 4-bit
(d) 4-bit + fp32 copy
(b) 4-bit frozen + LoRA
(a) bf16 + float copy
(b) start (open) to end (filled), one line per seed
×: (b) with its adapter merged into the 4-bit weights
GSM8K test accuracy (%)
4-bit weights only
needs 16-bit weights
50
55
60
65
70
75
(c)
ticker, 4-bit
(d)
4-bit + fp32 copy
(b)
4-bit frozen + LoRA
(a)
bf16 + float copy